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Water-MAS: A multi-agent LLM framework with instruction-data decoupling for smart water management

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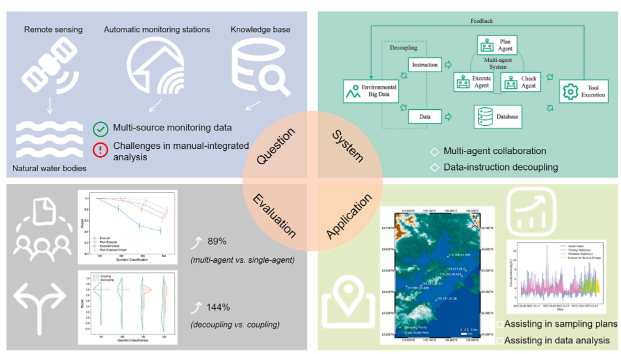
HIGHLIGHTS

- Proposed a multi-task water environment processing framework leveraging multi-agent collaboration and instruction-data decoupling to enhance system modularity and scalability.
- Built a water management evaluation dataset (260 knowledge-based questions and 114 task-oriented questions).
- Demonstrated performance improvements of multi-agent collaboration (89% relative gain) and instruction-data decoupling (143% relative gain).
- Validated the framework's real-world applicability in water sampling and water quality forecasting.

GRAPHIC ABSTRACT



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ABSTRACT

Sustainable water management demands transforming heterogeneous, long-sequence water data into timely, data-driven insights. While single-agent systems can perform basic tasks through tool integration, they suffer from low accuracy, limited adaptability, and insufficient big data analysis capabilities, rendering them unsuitable for complex, multi-step water management workflows. To overcome these constraints, we propose Water-MAS, a large language model (LLM)-based multi-agent framework designed for diverse water management tasks. The framework integrates three specialized agents (planning, execution, and checking) and introduces an instruction-data decoupling mechanism that separates task instructions from the underlying data, reducing computational overhead. Experiments demonstrate that, in four-tool invocation tasks, multi-agent collaboration achieves 89% higher recall (vs. single-agent baseline) under instruction-data decoupling, and the decoupling mechanism boosts recall by 143% relative to the coupled approach within the multi-agent framework. Analysis across foundation models reveals that smaller-scale models match performance of larger counterparts (Qwen3-8B vs. GPT-OSS-20B: F1 scores = 0.68 vs. 0.72), providing guidance for balancing system performance and model size. Case studies on water sampling and water quality forecasting validate the framework's real-world applicability. This work offers

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a reusable framework for deploying LLM-based multi-agent systems to enable automated tasks in environmental monitoring and data analysis.

1. Introduction

Adequate, safe water and sustainable water systems are essential for the survival and development of human societies. According to The Sustainable Development Goals Report 2025 (United Nations, 2025), water systems currently face significant pressures from pollution, water scarcity, and weak governance. Achieving sustainable water management requires effective analysis of water-environment data to support subsequent management and decision-making processes (Ahmed et al., 2024; Miller et al., 2025; Nong et al., 2025). Meanwhile, the types and methods of acquiring water environment data have become increasingly diverse. These include automated monitoring device data (Ruan et al., 2023; Wu et al., 2026), satellite remote sensing data (Wang et al., 2024c; Yan et al., 2025), model simulation data (Yan et al., 2024; Yang et al., 2025c), and community-sensed data (Dong et al., 2024; Tufail et al., 2025). However, due to the multiple data sources, high information density, and significant differences in how data is organized across these sources, manually interpreting and reorganizing this complex information is both labor-intensive and challenging. This manual approach severely limits the efficient utilization of data resources and results in pronounced "information silo" effects.

The emergence of large language models (LLMs) has opened new possibilities for automatic analysis and efficient utilization of multi-source heterogeneous water data. LLMs have been widely applied across multiple domains, including text mining and database construction in lifecycle assessment and toxicology (Kumar et al., 2025; Liang et al., 2024), decision-making recommendations during extreme natural disasters (Xie et al., 2025; Zhu et al., 2024), remote sensing image analysis (Bazi et al., 2024; Liu et al., 2025), and professional question-answering systems (Yang et al., 2025a; Zhang et al., 2025). However, sole LLMs remain fundamentally limited for operational water-environment workflows. They cannot natively query databases, call analytical routines, or trigger monitoring-related computations, and their outputs are bounded by the completeness and quality of the provided context (Xu et al., 2025). As a result, purely prompt-driven use hardly delivers auditable, reproducible results for compliance- and evidence-oriented applications.

Recently, LLM-based multi-agent systems have received increasing attention due to their superior robustness and high accuracy in complex scenarios, which combines the reliability and adaptability of multiple LLMs' collaboration with the interactive awareness capabilities of agents (Wang et al., 2024a; Xi et al., 2025). Multi-agent systems reduce individual agent burdens through architectural design, assigning clear task objectives to each agent to enable specialized focus and collaborative processing of complex tasks (Xu et al., 2024). Benefiting from this, multi-agent systems have been applied in diverse fields, including water resource optimization in distribution networks (Hu et al., 2023; Wang et al., 2026a), constructing multi-decision-level frameworks for large-scale farmland irrigation scheduling (Agyeman et al., 2025), and establishing multi-stakeholder game-theoretic models for ecological water resource governance (Meng, 2024). Among these applications, the EPANET-Agentic system integrates professional simulation tools and introduces a human-in-the-loop framework, ensuring industrial-grade reliability (Wang et al., 2026b). However, for water environment management requiring automatic multi-source data fusion and long-sequence data processing, developing a modular, reusable, and highly reliable architecture using multi-agent technology remains a significant challenge.

In this work, we develop a Water-MAS for automatic multi-source and long-sequence water data discovery and analysis with an instruction-data decoupling mechanism. The objectives of this study are to (i) construct a modular multi-agent water-environment data processing framework that separates instructions from underlying data representations, (ii) quantitatively evaluate the benefits of multi-agent collaboration for both knowledge-intensive reasoning and task-oriented tool execution in the water environment, (iii) assess the robustness and adaptability of the proposed architecture across nine open-source foundation models, and (iv) demonstrate practical utility through representative case studies, including compliant sampling plan generation and water-quality time-series forecasting. Collectively, this work establishes a reusable, collaborative paradigm for tasks in environmental monitoring and data analysis.

2. Method

2.1. System design

2.1.1. Multi-agent collaboration architecture design

The system is designed to provide accessible water-environment knowledge support for the public, while delivering actionable decision-making support for engineers (Fig. 1a). At its core lies a multi-agent architecture, in which three specialized agents coordinate task planning, execution, and validation through a structured workflow (Fig. 1b). This design enables each agent to focus on completing its designated tasks through specialized division of labor, thereby achieving optimal system performance. Moreover, this architectural approach is crucial for domains requiring high reliability, such as water environmental monitoring and management.

The detailed workflow is as follows: Upon receiving user input tasks, the planning agent first uses specific tools to decompose the task into multi-step executable procedures, then outputs the plan to the execution-checking iterative loop. In each iteration cycle, the execution agent selects the appropriate tool and its parameters based on the input mask message list, and then the tool executes them. The system is equipped with 14 tools, among which are sampling point generation, elevation data query, knowledge base retrieval, correlation analysis, and Long Short-Term Memory (LSTM) prediction (Tables S1–S2). The execution results are subsequently submitted to the checking Agent, which performs two critical checks: verifying whether the current step results match expectations, if inconsistencies are detected, it issues restart instructions to the execution agent along with next-step prompts; determining task completion status, if the task is successfully completed, the iteration process terminates immediately. The three agents are differentiated through distinct system prompts (Texts S1–S3) and varying input message lists to achieve their functional positioning.

2.1.2. Decoupled instruction-data mechanism design

Inspired by indexing methods in database systems (Mehravar et al., 2023; Wu et al., 2025), multi-agent systems can replace long data with concise indices, thereby allowing the system to focus on instruction execution and reduce the overhead of large-scale data processing in attention mechanisms. The decoupled instruction-data architecture enables reliable tool invocation on large water-environmental datasets under strict context-length constraints, which is a key component of Water-MAS. Instead of concatenating all prompts, tool calls and raw outputs into a single merged message list, decoupled instruction-data architecture maintain three coordinated message lists, a complete message list that stores full tool outputs as a persistent data layer, a masked message list that replaces large payloads with anonymous IDs and accumulates tool calls and guidance, and a summary message list that holds short LLM-readable summaries generated by a dedicated summary module (Fig. 1c and Text S4). A pair of translation layers map between anonymous IDs and full payloads during tool calls, allowing the planning, execution and checking agents to reason primarily over compact masked and summary representations while the tools operate on the original data. This decoupled design reduces context overhead, mitigates failures caused by large-scale parameter transmission, and provides stable, traceable references for complex multi-step water-data workflows (Further details are provided in Text S5).

2.1.3. *Architecture scalability*

The proposed architecture facilitates seamless extension of new tools through user configuration files that govern tool registration (Fig. S1). Specifically, during the main program execution, the system first reads a user configuration file (in JSON format) and subsequently registers the corresponding tools as dictated by the JSON file. This methodology eliminates the need for code modification when expanding tools and allows users to autonomously select which tools to activate.

Given that single-agent versus multi-agent architectures and data

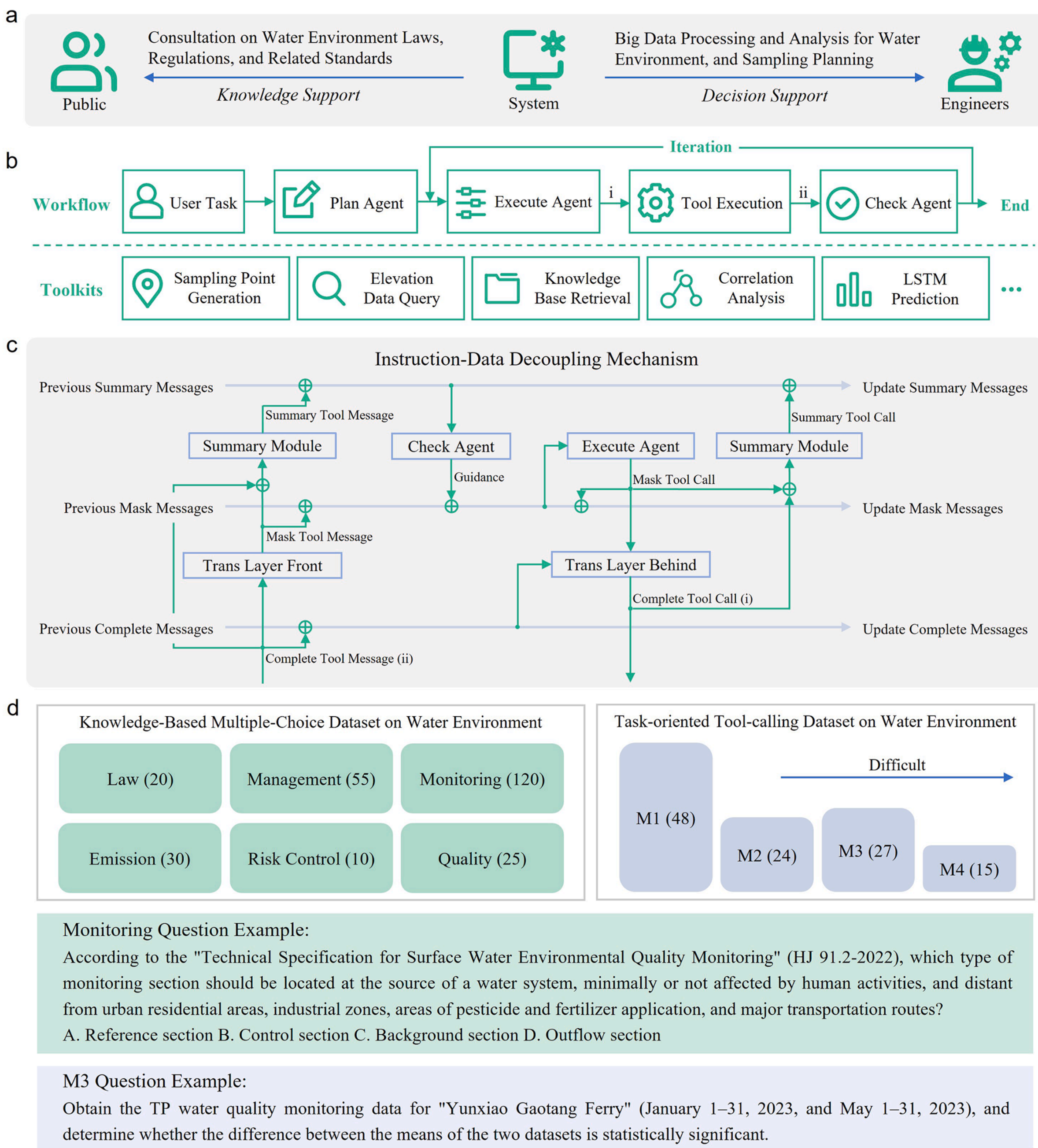

Fig. 1. Multi-agent system architecture design, instruction-data decoupling mechanism and datasets. **a.** System application scenarios, encompassing knowledge support for the public and decision support for engineers. **b.** The multi-agent system architecture design, encompassing the workflow from user task input to task execution completion. It illustrates the collaboration among the planning agent, execution agent, and checking agent, the iterative execute-check process, and depicts a subset of the tools available to the system. Annotations i and ii in (b) represent the Complete Tool Call and the Complete Tool Message, respectively, which are also shown in (c). **c.** The instruction-data decoupling mechanism, demonstrating the flow of messages within one iteration step, from the complete tool-invocation message input to the complete tool-execution result output. This primarily includes three message lists and the coordination among components. **d.** Two types of datasets: a knowledge-based multiple-choice dataset and a task-oriented tool-calling dataset, with the numbers in parentheses indicating the respective counts of questions in each category. M1, M2, M3, and M4 represent the minimum number of tools required for the task (used as an indicator of question complexity) being 1, 2, 3, and 4, respectively, with complexity increasing from M1 to M4.

coupling versus decoupling are suited to different scenarios (depending on task complexity and data requirements), an adaptive routing mechanism has been explored (Text S6 and Fig. S2). This mechanism implements an LLM-driven decision layer at the top of the architecture to execute different scripts based on the decision outcome.

Recognizing the critical role of anomaly detection in certain water scenarios, we provide an optional anomaly detection tool (Table S1). This tool can be configured to automatically trigger as a default subtask whenever a user issues a generalized query related to water quality evaluation, analysis, monitoring, or description.

2.2. Dataset

To comprehensively evaluate the system's performance on knowledge-based and task-based questions, we constructed a dataset comprising 260 knowledge-based multiple-choice questions and 114 task-based tool invocation questions (Fig. 1d; question examples are provided in Text S7). The 260 knowledge-based multiple-choice questions were developed based on 52 water environment-related laws, regulations, and industry standards (Table S3). Each question is a single-answer multiple-choice question with four options, and all questions have been manually verified. The 114 task-oriented tool-calling questions are categorized into four levels based on the number of required tools: the simplest type requiring only one tool call (M1, 48 questions), the simpler type requiring two tool calls (M2, 24 questions), the more complex type requiring three tool calls (M3, 27 questions), and the most complex type requiring four tool calls (M4, 15 questions). These task-oriented questions were designed based on real-world data collected from the following locations: Xiamen Bay, Meizhou Bay, Ningde Sanduao, and Yunxiao Gaotang Ferry Water Quality Monitoring Station (a non-tidal reach station). The Lianjiang Guantou Water Quality Monitoring Station (a tidal reach station) is mentioned here solely as an illustrative example (to show the performance of complicated hydrodynamic conditions) and its data were not included in the dataset. Each task-oriented tool-calling question has been manually categorized and paired with a human-verified reference tool-call sequence. Bathymetric data for the three bays were obtained from the GEBCO dataset (https://download.gebco.net/). The real-time monitoring data for the Yunxiao Gaotang Ferry Water Quality Monitoring Station and Lianjiang Guantou Water Quality Monitoring Station were sourced from the Real-time Data Release System for Automatic Monitoring of National Surface Water Quality (https://szzdjc.cnemc.cn:8070/GJZ/Business/Publish/Main.html).

2.3. Experimental setup

2.3.1. Experimental variable setup

2.3.1.1. Comparison of multi-agent collaboration strategies. The effectiveness of multi-agent collaboration was evaluated on both knowledge-based and task-oriented questions, with tests conducted under decoupled instruction-data architecture. For knowledge-based questions, the performance of five system configurations were tested on a dataset of 260 questions (using Qwen3-1.7B, Qwen3-4B, Qwen3-8B, Qwen3-14B as base model to compare different sizes): (i) a standalone LLM with internal knowledge base, (ii) a standalone execution agent, (iii) collaboration between a planning agent and an execution agent, (iv) collaboration between an execution agent and a checking agent, and (v) full collaboration involving a planning agent, an execution agent, and a checking agent. Each configuration was assessed using completion rate and accuracy. For task-oriented questions, four system configurations were evaluated on a test set of 114 questions, using Qwen3-8B as the base model (see Section 2.3.1.3 with other base models). These configurations (Fig. S3) were the same as those for knowledge-based questions, except for the standalone LLM (since it does not have the capability to call tools). To further demonstrate the capabilities of the multi-agent architecture, an all-in-one agent (prompt in Text S8) was also compared, and the results are presented in Fig. S4. Performance for each configuration was measured using precision, recall, and F1 score.

2.3.1.2. Comparison of instruction-data decoupling strategies. Using a test set of 114 task-oriented questions, the performance of the planning agent, execution agent, and checking agent system was evaluated under conditions of instruction-data coupling and instruction-data decoupling (using Qwen3-8B as base model, evaluation metrics include precision, recall, F1 score). Instruction-data coupling refers to the scenario where the planning agent, execution agent, and checking agent share a merged message list. In contrast, instruction-data decoupling divides the message list into three separate lists: the summary message list, the masked message list, and the complete message list.

2.3.1.3. Comparison of large language model selection. The performance of nine open-source LLMs was evaluated using the M4 question set (the most challenging subset of 114 task-oriented questions) and the same prompts (Texts S1–S3) within a collaborative system of planning, execution, and checking agents, to demonstrate their performance limits. The evaluation employed an instruction-data decoupling configuration, with performance metrics including precision, recall, F1 score, and token count. The nine open-source LLMs evaluated are Qwen3-1.7B, Qwen3-4B, Qwen3-8B, Qwen3-14B, Gemma-3-4B, Gemma-3-12B, Granite-4-H-tiny, Granite-3.2-8B, and GPT-OSS-20B (details provided in Table S4).

2.3.2. Experimental environment and hyperparameters

The code was developed using Python 3.12.10. All LLMs were invoked locally via LM Studio version 0.3.23. The Context Length for all LLMs was set to 10,240 tokens, with a Temperature of 0.1, Top K Sampling of 40, Repeat Penalty of 1.1, Min P Sampling of 0, and Top P Sampling of 0.95. Except for GPT-OSS-20B, which employed MXFP4 quantization, all other models utilized Q4_K_M quantization. For LLMs capable of both thinking and non-thinking modes, the non-thinking mode was adopted. To ensure reproducibility, a fixed random seed of 10 was used for any content requiring random generation within the tools. In multi-agent systems, all agents were based on the same underlying LLM.

2.4. Evaluation metrics

2.4.1. Evaluation metrics for knowledge-based questions

Completion rate and accuracy are employed to evaluate knowledge-based questions. Due to the relatively complex system architecture, the completion rate is used to measure the proportion of questions that the system successfully completes without errors. The formula for calculating the completion rate for a set of questions can be expressed as:

$$\text{Completion} = 1 - \frac{\sum_{i=1}^{n} e_i}{n} \tag{1}$$

where e_i equals 1 if an error occurs on the ith question and 0 otherwise; n denotes the total number of questions.

Accuracy is defined as the proportion of system responses that match the answers verified by human review. The formula for calculating the accuracy for a set of questions is as follows:

$$\text{Accuracy} = \frac{\sum_{i=1}^{n} c_i}{n} \tag{2}$$

where c_i equals 1 if the answer to the ith question is correct, and 0 otherwise.

2.4.2. Evaluation metrics for task-based questions

Precision, recall, and F1 score are adopted to evaluate task-oriented questions. Specifically, precision measures the proportion of the system's tool invocations that match the reference tool calls. A higher value indicates a greater rate of correct tool selection or accurate input parameter generation. The average precision across a set of questions is calculated as follows:

$$\text{Precision} = \frac{\sum_{i=1}^{n} \frac{|SR_i \cap SS_i|}{|SS_i|}}{n} \tag{3}$$

where SR_i denotes the set of reference tool calls for the ith question, SS_i denotes the set of tool calls generated by the system for the ith question, and n is the total number of questions.

Recall measures the extent to which the system's tool calls cover the reference tool calls. A higher recall value indicates that the system can more comprehensively encompass the reference outcomes. This metric is particularly critical in high-reliability domains such as water environment management. The average recall across a set of questions is calculated as follows:

$$\text{Recall} = \frac{\sum_{i=1}^{n} \frac{|SR_i \cap SS_i|}{|SR_i|}}{n} \tag{4}$$

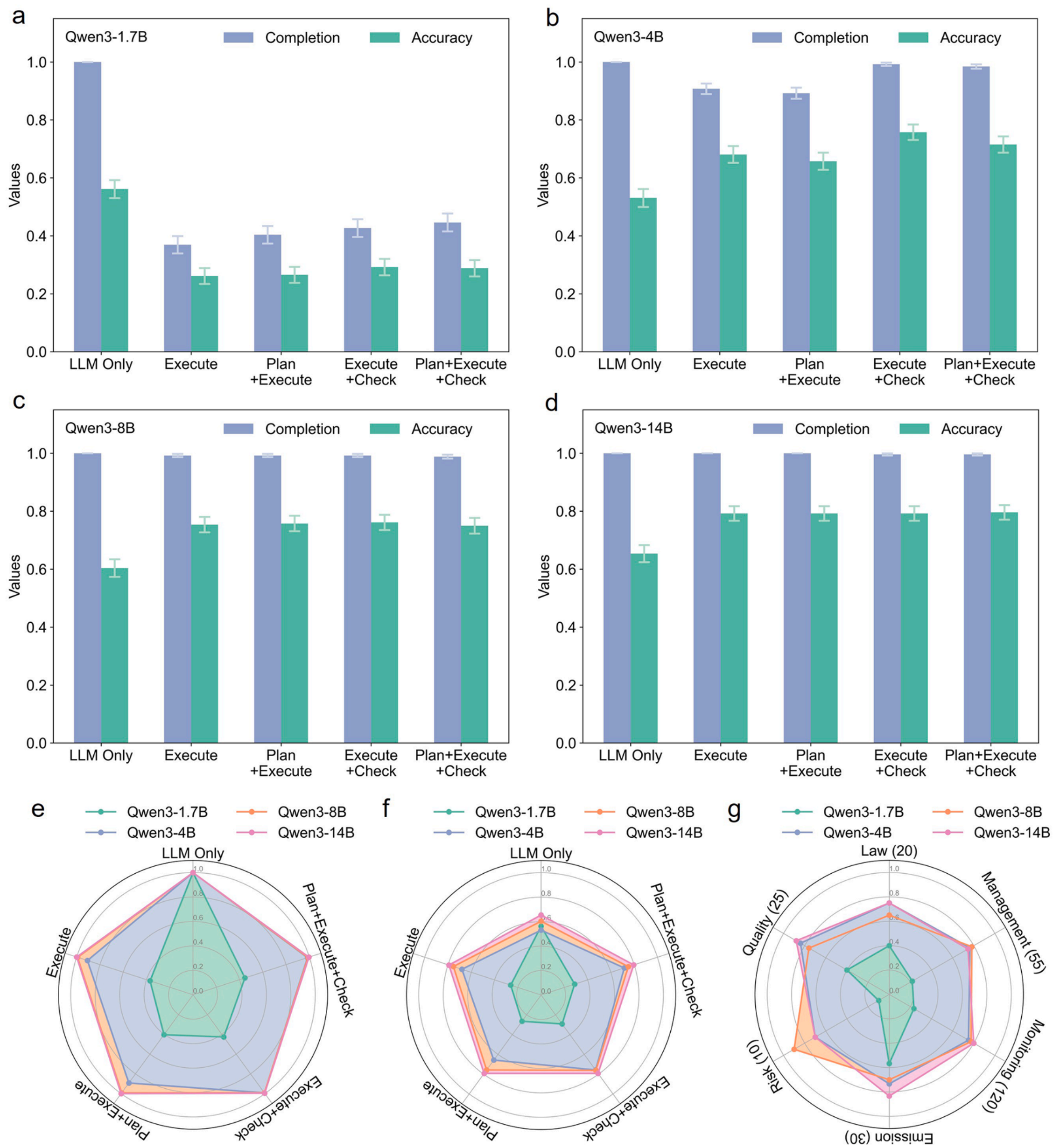

Fig. 2. Response of different architectural systems to knowledge-based multiple-choice questions. **a.** Completion rate and accuracy using Qwen3-1.7B as the base model; **b.** Completion rate and accuracy using Qwen3-4B as the base model; **c.** Completion rate and accuracy using Qwen3-8B as the base model; **d.** Completion rate and accuracy using Qwen3-14B as the base model; **e.** Comparison of completion rates across base models with different parameter scales; **f.** Comparison of accuracy across base models with different parameter scales; **g.** Comparison of accuracy across base models with different parameter scales according to subject categories. "LLM Only" indicates reliance solely on the LLM's internal knowledge acquired during its pre-training; "Plan" represents the planning agent; "Execute" represents the execution agent, which has the ability to invoke tools; "Check" represents the checking agent. The completion rate refers to the proportion of test questions for which the system executed successfully without errors out of the total number of test questions; the accuracy refers to the proportion of test questions for which the system provided correct answers out of the total number of test questions. Error bars represent the standard error of the mean (SEM) for the respective group.

The F1 score is a balanced metric that combines precision and recall, providing a more comprehensive reflection of system performance. Its calculation formula is as follows:

$$\text{F1score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (5)$$

2.4.3. Statistically analysis

For comparisons involving more than two groups, Welch's ANOVA was initially employed to assess overall significant differences, as this method does not assume homogeneity of variances. If the Welch's ANOVA result was statistically significant ($P < 0.05$), post hoc pairwise comparisons were conducted to identify specific group differences. All pairwise comparisons utilized the Mann–Whitney U test, with P values adjusted using the Benjamini/Hochberg (FDR) method to control for the false discovery rate associated with multiple comparisons. For direct comparisons involving only two groups, the Mann–Whitney U test was also applied to evaluate statistical significance.

3. Result and discussion

3.1. Multi-agent collaboration

3.1.1. Multi-agent collaboration for enhancing water environment understanding

Compared to general knowledge, the water environment domain contains extensive proprietary knowledge, such as definitions of technical terms and specific mechanism (Chen et al., 2024; Wei et al., 2024). Water-MAS is designed for the water environment domain, where addressing encountered challenges requires a certain level of subject matter expertise. Therefore, knowledge-based multiple-choice questions were developed to evaluate the system's effectiveness. To assess the impact of multi-agent collaboration on knowledge question performance, experiments were conducted across five configurations: (i) relying solely on the LLM's internal knowledge base, (ii) execution agent only, (iii) planning and execution agents, (iv) execution and checking agents, and (v) planning, execution, and checking agents (Fig. 2, significance analysis in Fig. S5 and Table S5). Based on the Qwen3-4B foundation model (which balances parameter size and performance in the scenario), different agent architectures were evaluated. Adding an execution agent improved accuracy by 28% compared to relying solely on the LLM's knowledge base (increasing from 0.53 to 0.68, $P < 0.01$), demonstrating that external tools significantly enhance result accuracy and validating the effectiveness of Retrieval-Augmented Generation. However, introducing a planning agent reduced system effectiveness, with mean accuracy dropping from 0.68 to 0.66. This likely resulted from increased system complexity, as this task is inherently single-step and the execution agent alone could correctly invoke tools. This hypothesis is supported by the decreased completion rate (from 0.91 to 0.89). Conversely, integrating a checking agent improved performance, raising accuracy from 0.68 to 0.76 (12%, $P < 0.1$). This indicates effective collaboration between execution and checking agents, where the checking agent identified errors under execution and guided improvements. From the perspective of computational costs, for simple query tasks, employing either a standalone execution agent or a system where the execution agent collaborates with the checking agent could be appropriate. An execution agent is more economical in terms of token usage and improves computational efficiency. In contrast, a system where the execution and checking agents collaborate offers higher accuracy but requires more tokens. This trade-off should be balanced based on practical application requirements.

When comparing models of different parameter scales, the 1.7B-parameter model was too limited in capacity to correctly invoke tools for such tasks, resulting in consistently low completion rates and accuracy across all system architectures (Completion Rate < 0.5, Accuracy < 0.3). In contrast, a 4B-parameter model proved optimal when paired with appropriate architecture (execution and checking agents, Completion Rate $= 0.99$, Accuracy $= 0.76$). For models with 8B or more parameters, system performance reached saturation, with all architectures achieving comparable results (Completion Rate = 0.99–1.00, Accuracy = 0.75–0.80). These observations align with findings from a parallel study on hydrological question-answering systems (Kizilkaya et al., 2025), which similarly reported inferior performance in smaller models and diminishing returns beyond a certain parameter threshold. The exploration of parameter scaling provides practical technical guidance for deploying similar systems in water environment applications, demonstrating that while model scale significantly impacts performance for knowledge-intensive tasks, gains plateau once a critical capacity is reached.

When examining different categories within the water environmental field (Fig. 2g, significance analysis in Table S6), no significant differences were observed among the 4B, 8B, and 14B parameter LLMs ($P > 0.1$), indicating that the system demonstrates consistent performance across water environment-related laws, management standards, monitoring standards, pollutant emission standards, risk control standards, and quality standards. This consistency reflects the system's broad-spectrum adaptability to knowledge-based questions in the water environment domain.

3.1.2. Multi-agent collaboration for promoting water environment task execution

To evaluate the impact of multi-agent collaboration on the system's performance in four categories of task-oriented questions (M1–M4, representing the minimum number of tools required for the task), ablation experiments were conducted and divided into four groups: execution agent only, planning and execution agents, execution and checking agents, and planning, execution, and checking agents (Fig. 3, significance analysis in Tables S7–S14). In terms of recall, all groups performed perfectly on the simplest M1 questions. As question difficulty increased, all groups showed declining performance with widening performance gaps between groups. For the most challenging M4 questions, adding a planning agent, adding a checking agent, and three-agent collaboration improved mean recall by 55% (0.643, $P < 0.05$), 71% (0.707, $P < 0.01$), and 89% (0.782, $P < 0.01$) respectively compared to the execution-only agent group (0.414), demonstrating the comprehensive tool-calling capability of multi-agent collaboration in complex tasks. Among the three multi-agent systems, minimal differences were observed in simpler M2 questions (Recall = 0.96–1, $P > 0.1$), while in more difficult M3 questions, three-agent collaboration outperformed two-agent collaboration (Recall = 0.96 vs. 0.81–0.89, $P < 0.1$). Notably, the three-agent collaboration achieved recall of 0.78–1 across all tested question types. The comparative analysis across question difficulties reveals greater benefits from multi-agent systems in complex tasks (Hu et al., 2024; Jing et al., 2025). This improvement is attributed to multi-agent systems distributing cognitive loads through role specialization, allowing individual agents to focus on specific responsibilities while overcoming the limitations of single-agent architectures that struggle to simultaneously handle planning and execution in complex scenarios.

In the precision metric, for M3 questions, the group with both planning and execution agents demonstrated higher precision compared to the group with only execution agent (Precision = 0.69 vs. 0.50, $P < 0.05$). This was because the planning agent broke down complete tasks into subsequent tasks and provided guidance for subsequent task execution processes (Shen et al., 2024), reducing incorrect tool calls and improving system efficiency. Notably, this performance improvement from introducing the planning agent contrasts with knowledge-based multiple-choice questions (where the planning agent reduced overall accuracy by 3–5%, as shown in Section 3.1.1). The reason lies in the increased number of execution steps required for complex tasks (M3, M4), where the planning agent's clear guidance enhances the execution agent's validity. Further analysis of the impacts of planning and checking

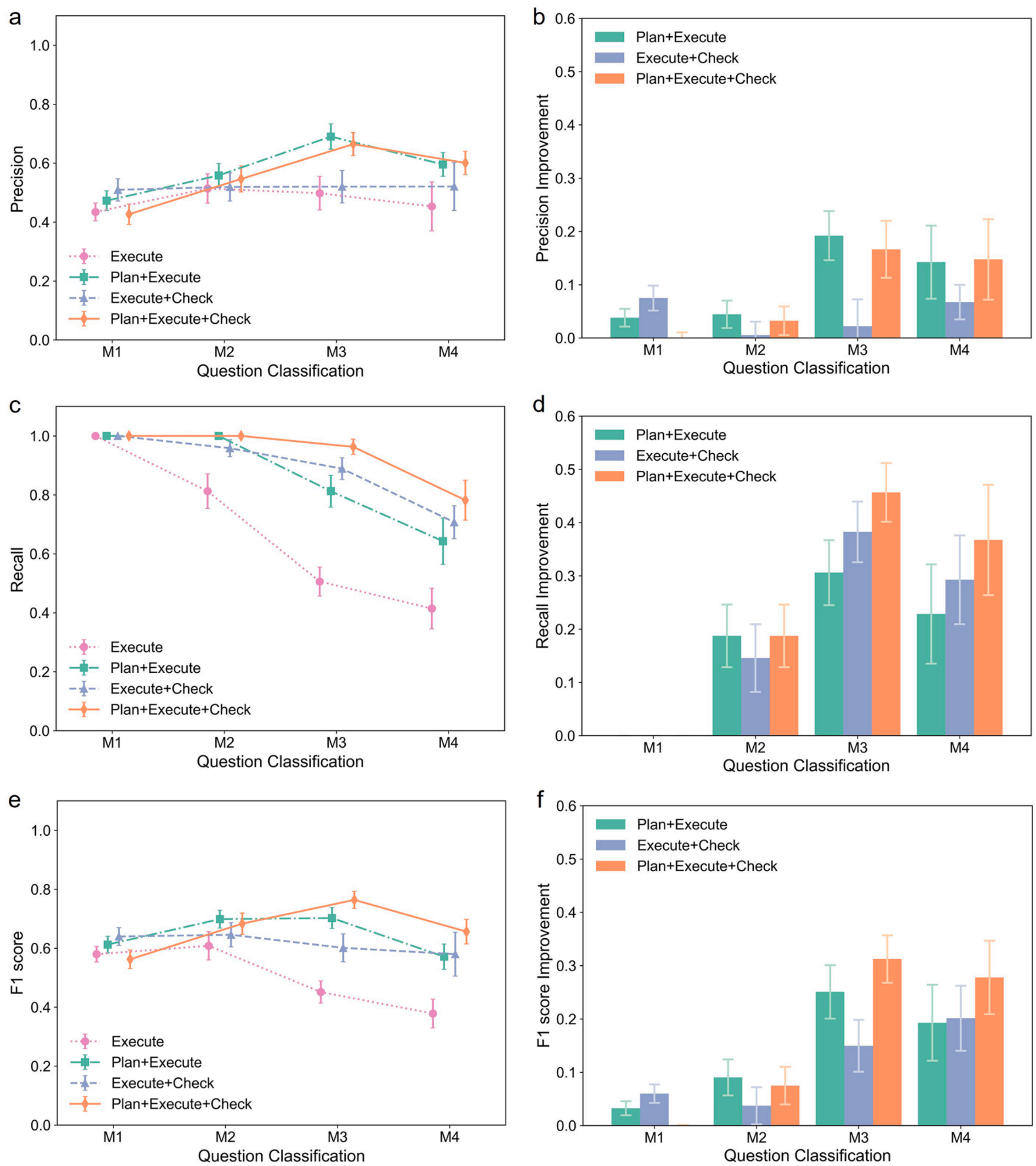

Fig. 3. Response of different architectural systems to task-oriented tool-calling questions. **a.** Precision; **b.** The absolute improvement in precision for other groups relative to the standalone execution agent group; **c.** Recall; **d.** The absolute improvement in recall for other groups relative to the standalone execution agent group; **e.** F1 score; **f.** The absolute improvement in F1 score for other groups relative to the standalone execution agent group. M1, M2, M3, and M4 represent the minimum number of tools required for the task (used as an indicator of question complexity) being 1, 2, 3, and 4, respectively, with complexity increasing from M1 to M4. "Plan" represents the planning agent; "Execute" represents the execution agent; "Check" represents the checking agent. Error bars represent the standard error of the mean (SEM) for the respective group.

agents on system performance revealed that the planning agent contributed more to precision improvement, while the checking agent had a greater positive effect on recall. Specifically, the planning agent increased the initial validity of tool calls during task execution, whereas the checking agent improved overall task execution validity through iterative corrections. This viewpoint can also be supported from iteration steps (Fig. S6), for M4 questions, the group of checking and execution agents required more execution steps (Step = 6.7) than the group of planning and execution agents (Step = 4.1). Therefore, integrating planning, checking, and execution agents led to substantial improvements in both precision and recall. However, if the planning agent fails to generate the expected correct plan, it may reduce the accuracy of subsequent execution. To further refine the framework, incorporating the planning agent into an iterative loop could be a promising direction.

In the comprehensive F1 score metric, this perspective was further validated. For the most complex M4 questions, a three-tier performance pattern emerged, the best performance was achieved by the three-agent collaboration (F1 score = 0.68), followed by two-agent collaborations (F1 score = 0.60–0.62), and the lowest performance came from single-agent systems (F1 score = 0.43). This clearly demonstrated the advantages of multi-agent systems over single-agent systems in handling complex tasks.

3.2. Instruction-data decoupling for processing long-sequence water environment data

To investigate the mitigation effect of the instruction-data decoupling mechanism on the context length limitations of LLMs, the performance of both the instruction-data coupling and decoupling mechanisms was evaluated on four categories of task-oriented questions (Fig. 4, significance analysis in Tables S15–S17). In the recall metric, the decoupling mechanism achieved higher mean recall across all question difficulties compared to the coupling mechanism, demonstrating superior answer completeness. Notably, the performance gap widened with increasing question complexity. For M1 questions, the decoupling mechanism showed only a 9% improvement over the coupling mechanism (0.92 vs. 1.00, $P < 0.05$). However, for M4 questions, the decoupling mechanism exhibited a 143% improvement (0.32 vs. 0.78, $P < 0.01$). These results highlight the instruction-data decoupling mechanism's especially significant advantages in complex tasks and validate its necessity for multi-step tasks involving large-scale data.

In the precision metric, the decoupling mechanism first increases and then decreases with increasing question difficulty. This trend may stem from two factors: first, LLMs become overly sensitive to simple tasks, triggering unnecessary tool calls that lower precision (e.g., precision of 0.43 for M1 questions with the decoupling mechanism); second, lower correct tool invocation rates for complex tasks, which further reduces system precision in handling such tasks (e.g., precision of 0.66 for M3 questions and 0.6 for M4 questions with the decoupling mechanism). In contrast, for the coupling mechanism, performance across tasks M1 to M4 exhibits a declining trend, primarily attributable to reduced accuracy in tool invocation under increased task complexity. Comparative analysis of the two mechanisms reveals that the decoupling mechanism maintains a statistically significant advantage over the coupling mechanism in handling complex tasks (M3: Precision = 0.66 vs. 0.41, $P < 0.01$; M4: Precision = 0.6 vs. 0.21, $P < 0.01$), highlighting the importance of coordination and communication between agents (Sun et al., 2025; Yang et al., 2025b).

In the composite evaluation metric (F1 score), the decoupling mechanism shows a similar upward-then-downward trend due to the influence of precision. As shown in the distribution trend of F1 scores, the proportion of completely erroneous questions in the coupling mechanism gradually increases with question complexity (from M1 to M4), exhibiting a bimodal distribution pattern. This indicates poorer stability of the coupling mechanism in complex scenarios. Experimental results demonstrate that for simple tasks, the performance difference between coupling and decoupling mechanisms is minimal. Considering system complexity, adopting the coupling mechanism for straightforward tasks may be more efficient. However, for complex tasks, the decoupling mechanism significantly outperforms the coupling mechanism (M3: F1 score = 0.79 vs. 0.47, $P < 0.01$; M4: F1 score = 0.68 vs. 0.26, $P < 0.01$), making the instruction-data decoupling approach the more suitable choice for intricate tasks. Due to the increased difficulty in handling longer sequences, the results illustrate that the coupling mechanism is suitable for shorter sequences, while the decoupling mechanism is more appropriate for longer sequences, which are common in the water environment.

3.3. Different large language models for verifying systematic generalization

To evaluate the generalization ability of the system across different large language models, several open-source models were tested using the relatively challenging M4 dataset (Fig. 5, Table S18, significance analysis in Fig. S7–S9). In terms of recall, GPT-OSS-20B demonstrated

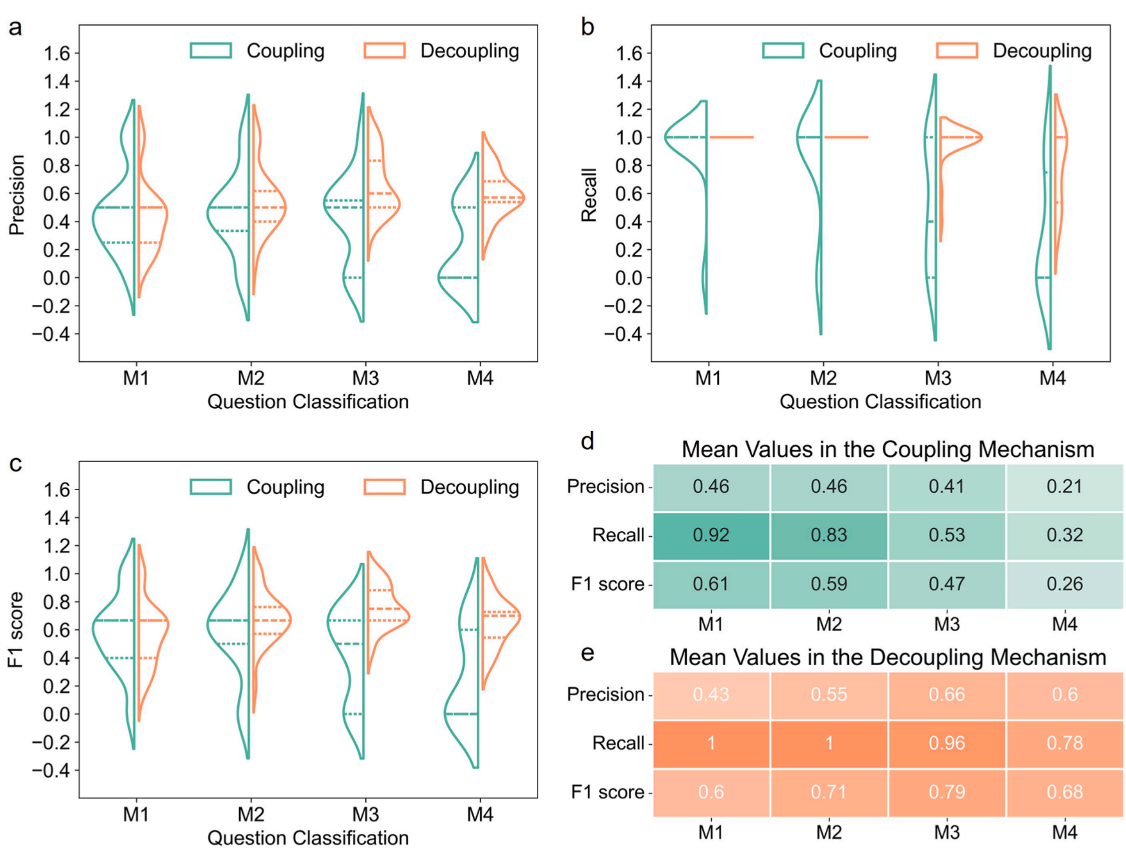

Fig. 4. Comparison between not adopting and adopting the instruction-data decoupling mechanism. **a.** Precision; **b.** Recall; **c.** F1 score; **d.** Mean values in coupling mechanism; **e.** Mean values in decoupling mechanism. M1, M2, M3, and M4 represent the minimum number of tools required for the task (used as an indicator of question complexity) being 1, 2, 3, and 4, respectively, with complexity increasing from M1 to M4. The dashed lines in the figure denote the quartiles of each group.

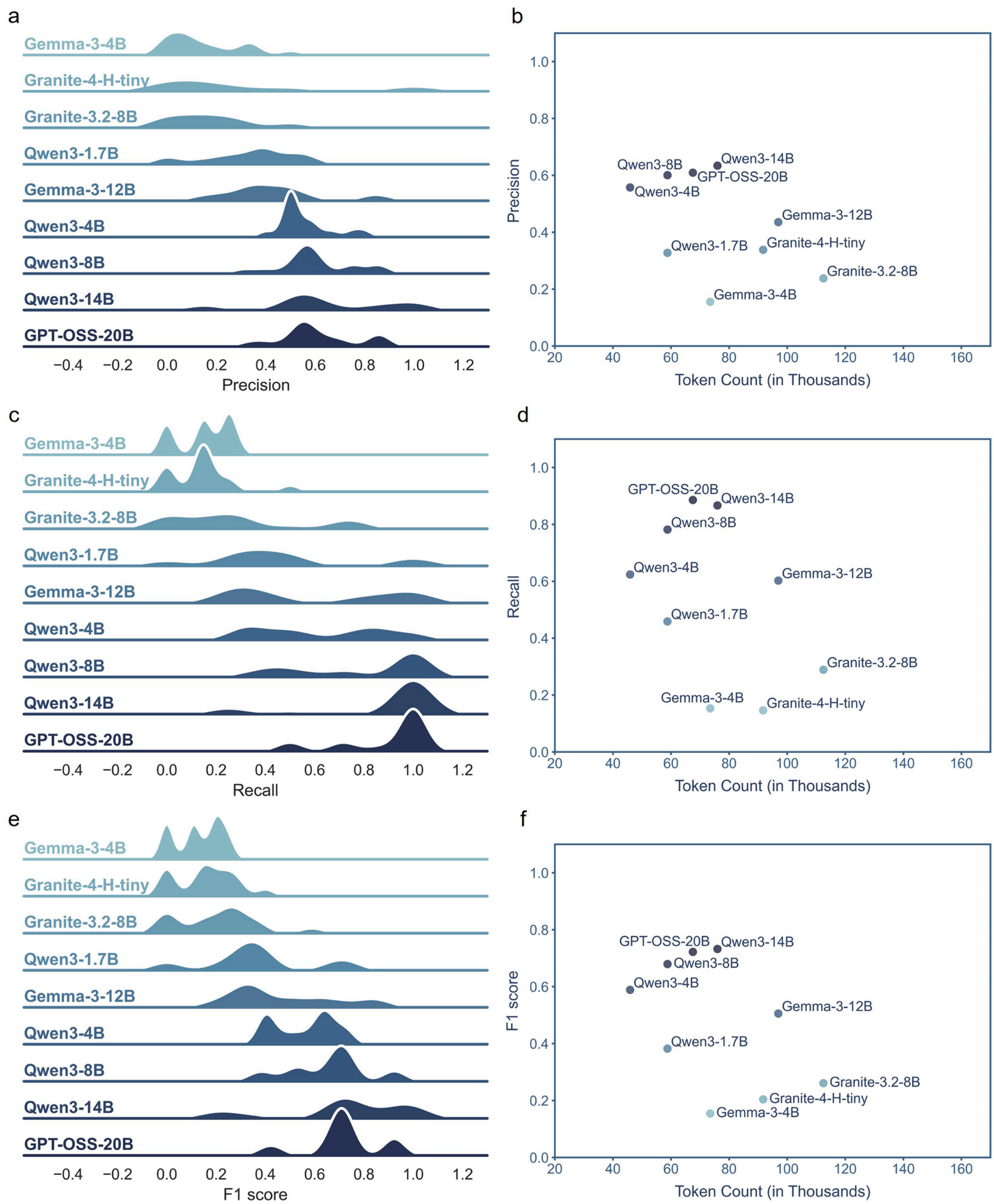

Fig. 5. Response of different base models within the system. **a.** Density distribution of Precision; **b.** Precision vs. token count (in thousands); **c.** Density distribution of Recall; **d.** Recall vs. token count (in thousands); **e.** Density distribution of F1 score; **f.** F1 score vs. token count (in thousands). In density distribution plots, the models are arranged in ascending order based on their mean F1 score from top to bottom. The test set used for this figure is the M4 questions (i.e., questions requiring a minimum of 4 tools to be invoked, which is the most challenging subset among all test sets). Token count (in thousands) refers to the average across instances of the sum of input and output tokens in a specific LLM.

the best performance, followed by Qwen3-14B and Qwen3-8B, achieving recall of 0.78–0.89, which indicates a practically usable level. When comparing models of similar parameter scales, Qwen3-4B (Recall = 0.62) showed significant improvement over Gemma-3-4B (Recall = 0.15), and Qwen3-8B (Recall = 0.78) outperformed Granite-3.2–8B (Recall = 0.29) and Granite-4-H-Tiny (Recall = 0.15), with statistical significance ($P < 0.01$). This highlights the superior performance of the Qwen3 series in these tasks. Additionally, DeepSeek-R1-Distill-Llama-8B (Recall = 0.17) and DeepSeek-R1-Distill-Qwen-7B (Recall = 0.18) were evaluated but excluded from the figures due to their thinking mode. Their low recall may be attributed to the incompatibility between the models' native output formats and the system's input requirements. Considering token efficiency, GPT-OSS-20B, Qwen3-8B, and Qwen3-4B achieved better results with fewer tokens. For precision, GPT-OSS-20B, Qwen3-14B, and Qwen3-8B performed well with precision of 0.6-0.63, indicating effective tool calls and fewer rejections by the checking agent against the execution agent. Consequently, these models achieved high precision with lower token consumption. The F1 score rankings aligned with the precision results, where GPT-OSS-20B (F1 score = 0.72), Qwen3-14B (F1 score = 0.73), and Qwen3-8B (F1 score = 0.68) demonstrated the best comprehensive performance.

This experiment demonstrates the system architecture's generalization capability across diverse models and provides reference for selecting foundational models in future deployments. The results also reveal that in specialized domains like water environment tasks, smaller-parameter models can achieve performance comparable to larger ones (e.g., Qwen3-8B vs. GPT-OSS-20B: F1 scores = 0.68 vs. 0.72, $P > 0.1$) (Kalyuzhnaya et al., 2025). This finding suggests potential cost reductions for deploying large language models in specialized domains (e.

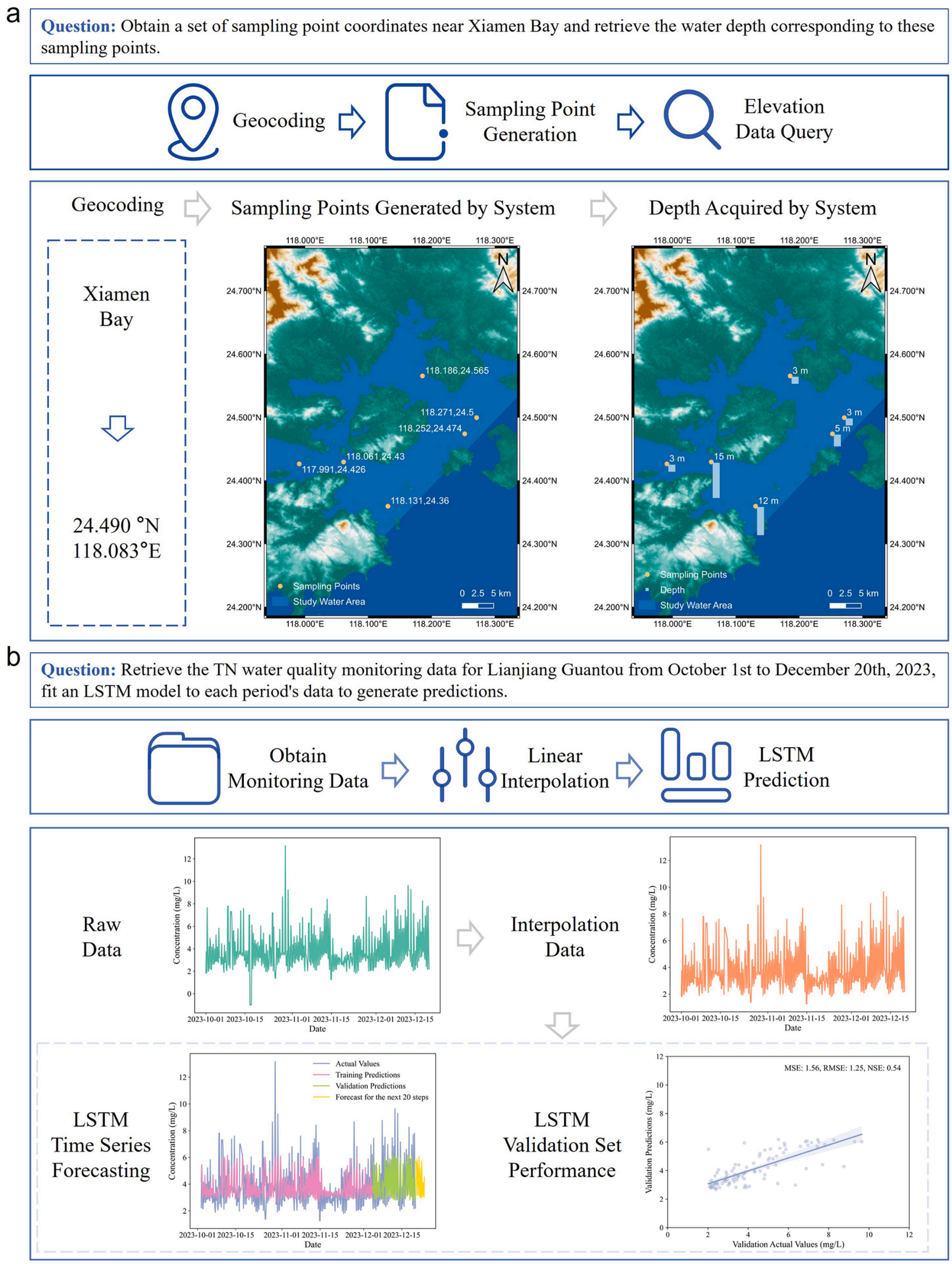

Fig. 6. Case studies, each of which consists of three components: the input real-world question, a system execution flowchart, and a visualization of the system execution steps. **a.** The system performs a water environment sampling instruction generation task. The map shown in the execution step visualization was created using QGIS 3.40.8, with the base map data derived from ASTER GDEM V3 (provided by the Geospatial Data Cloud site, Computer Network Information Center, Chinese Academy of Sciences; http://www.gscloud.cn). **b.** The system carries out a task involving time-series data analysis and prediction of pollutants. The execution step visualization for this case was generated using Python 3.12.10.

g., the water environment) and provides technical guidance for balancing system performance and model scale.

3.4. Case study

3.4.1. Water sampling task in environmental monitoring

Although water sampling is a common and important practice in environmental monitoring (Wang et al., 2024d), many sampling plans are still formulated manually. This task demonstrates the system's capability in generating instructions for water environment sampling (Fig. 6a), which can show a practical scenario for environmental monitoring that reduces labor intensity through automation. The system first employed the planning agent to devise a step-by-step strategy. During the iterative execution and validation phase, an execution agent followed the plan. It first called the geocoding tool to acquire the latitude and longitude coordinates of the target location. Subsequently, using a sampling point generation tool that integrated water extent data from local digital elevation model, it automatically generated latitude and longitude coordinates for six sampling points within the target area through grid-based random sampling. Finally, the water depth retrieval tool was employed to obtain the bathymetric data for each sampling point. The system successfully accomplished its predefined objectives, demonstrating its ability to automate certain aspects of instruction generation for water environment sampling tasks.

3.4.2. Water quality forecasting in environmental data analysis contexts

While predicting water quality at given points is a cornerstone of smart water management (Shang et al., 2025), the process of data collection, cleaning, and model calibration remains a tedious burden for engineers. This task demonstrates a specific case of the system addressing a monthly time-series analysis and prediction task for pollutant data (Fig. 6b), highlighting its potential to automatically address the growing demand for water data analysis. The system first employed the planning agent to decompose the overall task into sequential steps, followed by an iterative execution and validation phase. Initially, the system invoked a data retrieval tool, correctly passing relevant parameters to accurately extract the required water quality monitoring data from the database. Subsequently, the system utilized a linear interpolation tool to fill in the missing values, thereby generating a complete dataset suitable for further analysis and forecasting. Finally, the system activated the LSTM tool, which trained the model on the interpolated data sequence. The model achieved an acceptable result (MSE = 1.56, RMSE = 1.25, NSE = 0.54) on the validation set, then proceeded to predict the subsequent 20 steps. This automated process of task planning and execution aligned with expectations, demonstrating the system's capability in handling long-sequence water quality data analysis tasks.

Beyond two specific tasks, Water-MAS provides a reusable framework for handling heterogeneous (geographic information and water quality) and long-sequence (time series spanning a month with a 4-hour monitoring interval) data in water management and other similar domains, leveraging a synergistic integration of multi-agent collaboration and the instruction-data decoupling mechanism. Moreover, its extensible design allows the framework to incorporate specialized modules, such as online water monitoring (Wang et al., 2024b), emergency water management approaches (Carbon, 2024), and other prediction models (Li et al., 2024), supporting adaptation to diverse scenarios. In the future, the framework could become a key driver for automated, data-driven water management, extending beyond environmental monitoring and data analysis to a wide range of other scenarios, such as automatic analysis in data science, data fusion in geoscience, and high-reliability evaluation in life sciences.

4. Conclusion

This study focuses on environmental monitoring and data analysis in the water environment and constructs an automatic multi-agent system framework. By introducing an instruction-data decoupling mechanism, the framework demonstrates effective handling of complex and large-scale data questions. The following conclusions were drawn:

Multi-agent collaboration significantly enhances the system's effectiveness in handling complex tasks. In knowledge-based questions, the system combining execution and checking agents achieved approximately 12% improvement in accuracy compared to the single execution agent system ($P < 0.1$). For the most challenging M4 task-oriented questions, the three-agent collaboration (planning, execution, and checking) increased recall by 89% compared to the execution agent only group ($P < 0.01$).

The instruction-data decoupling mechanism shows obvious effectiveness in addressing large-scale data challenges. For the most difficult M4 task-oriented questions, the group implementing the decoupling mechanism showed a 143% improvement in recall compared to the coupled group ($P < 0.01$).

GPT-OSS-20B, Qwen3-14B, and Qwen3-8B demonstrate high compatibility with the three-agent collaborative framework. Among nine open-source LLMs tested on the most challenging M4 task-oriented questions, these three models achieved recall scores of 0.78–0.89, significantly outperforming other models (Recall = 0.15–0.62).

Case studies illustrate the system's capabilities in water environment sampling guidance generation and monthly water quality time-series forecasting, highlighting its practical potential in water management.

Declaration of generative AI in the writing process

Any use of generative AI in this manuscript adheres to ethical guidelines for use and acknowledgement of generative AI in academic research. Each author has made a substantial contribution to the work, which has been thoroughly vetted for accuracy, and assumes responsibility for the integrity of their contributions.

CRediT authorship contribution statement

Fangkun Lin: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Zhipeng Luo:** Visualization, Investigation, Formal analysis. **Binyu Ma:** Visualization, Formal analysis. **Zhiyu Zhang:** Validation, Data curation. **Kaiyan Liu:** Investigation, Formal analysis. **Xin Yu:** Supervision, Funding acquisition. **Boyan Xu:** Writing – review & editing, Project administration, Funding acquisition, Conceptualization. **Huabin Zeng:** Writing – review & editing, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.watres.2026.126163.

Data availability

Data will be made available on request.

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